SANJEY MANIAN

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SUMMARY

Computer science undergraduate (UT Dallas, 2028) who builds LLM-powered and machine learning systems in Python, including a Retrieval-Augmented Generation (RAG) system on Supabase and an AI player for a real-time multiplayer game. Stress-tested a multimodal generative AI model to log 100+ failures and builds enterprise data pipelines at NGAI. Looking to ship scalable AI features as Copart's Data & AI intern.

EDUCATION

B.S. in Computer Science - The University of Texas at Dallas Expected 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Databases (SQL), Linux/Unix Systems

TECHNICAL SKILLS

AI & LLM: Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), Conversational AI, Model Evaluation, Open-Source Models (NVIDIA Nemotron), Machine Learning (scikit-learn, XGBoost)
Languages & Data: Python (pandas), SQL (PostgreSQL), JavaScript, TypeScript, Java, C++
Full-Stack & DevOps: React, Node.js, Socket.IO (WebSockets), REST APIs, Docker, Git, Google Cloud (OCR), Supabase, Streamlit, Linux/Unix
Core Strengths: Data Structures & Algorithms, Debugging & Root-Cause Analysis, technical documentation, teamwork, continuous learning
Certifications: NOCTI Certification - Cybersecurity; AI Engineering with Python (in progress)

PROFESSIONAL EXPERIENCE

Undergraduate Data Science Researcher - The Data Mine (Purdue University) Jan 2026 - Present

- Built RAG retrieval on Supabase, as measured by faster complex-query retrieval speed, by modeling the schema and integrating Retrieval-Augmented Generation (RAG) into the data layer.
- Cut data-collection time by an estimated 80%, as measured by pipeline runtime vs. manual workflows, by engineering Python ETL pipelines for unstructured SEC.gov, PatentsView, and SAM.gov data for Air Force and DoD projects.
- Shipped a containerized Streamlit dashboard for large-scale commercial and defense datasets, as measured by a live app, by packaging the pipeline and frontend in Docker for scalable, consistent deployment.
- Analyzed 1,000+ records from PitchBook, USASpending.gov, and SAM.gov to surface commercial technology trends, as measured by dataset coverage, by writing technical reports for stakeholders.

EAA Data Engineer Intern - NGAI (Remote) Oct 2026 - Present

- Automated ingestion pipelines and REST API integrations for the Enterprise Advanced Analytics (EAA) platform, as measured by automated data flow, by building its Python and SQL foundations.
- Enforced enterprise data quality, as measured by automated validation on incoming records, by building normalization, entity resolution, and validation workflows in Python, SQL (PostgreSQL), and Frappe Framework.
- Governed reporting data end to end, as measured by lineage tracking and multi-tenant access controls, by implementing semantic models.

AI Trainer & Data Annotator - Handshake AI (Project Lizard) Jan 2026 - Present

- Found 100+ LLM failures and hallucinations, as measured by unique cases logged, by stress-testing a multimodal generative AI model on visual-reasoning prompts.
- Evaluated model quality on flagged edge cases, as measured by engineering retraining outcomes, by tracing root causes and writing training data.

Engineer / Researcher - Sports Analytics Club at UTD Aug 2026 - Present

- Modeled NHL shot quality in real time with a custom Expected Goals (xG) model, as measured by per-shot xG estimates, by engineering features from shot location, angle, distance, type, rebounds, and rush status.
- Quantified goaltender performance with Goals Saved Above Expected (GSAx), as measured by shot-map and goalie-analysis visuals, by training pandas, scikit-learn, and XGBoost models and charting in Streamlit and Plotly.

Events / Video Editor Officer - Google Developer Group UTD Aug 2026 - Present

- Led workshops and hackathons for the campus developer community, as measured by events delivered, by planning logistics.

Barista - Feng Cha Jun 2026 - Present

- Held 100% order accuracy across 150+ transactions per shift, as measured by POS audit logs, by double-checking every ticket.

PROJECTS

aMuse-Us - Multiplayer AI Social-Deduction Game HackTX, 2026

- Co-built a real-time game with a conversational AI player, as measured by 44 commits across 4 people, by developing a React/TypeScript frontend, a Node/Socket.IO server, and the AI player.

MEMEoTron - AI-Driven Meme Intelligence for Marketing Research HackUTD, Nov 2025

- Shipped real-time trend analysis in 24 hours, as measured by a live demo, by chaining the open NVIDIA Nemotron-Nano-4B-v1.1 model, Google Cloud OCR, and web scraping into a multimodal pipeline in Google Colab.

Automated Defense Intelligence Pipeline Small Business Consulting Corp. & The Data Mine

- Expanded database coverage across 5 intelligence sources, as measured by integrated registries, by building Python ETL pipelines.

Backpack Robot - User-Following & Obstacle-Evasion Robot Personal Project, Jul 2024

- Navigated obstacles autonomously, as measured by real-world trials, by programming an Arduino robot with ultrasonic sensors.